10.1.4 Applications of adsorption

Though we have innumerable applications for adsorption, we consider few of them

1. Gas masks: During world war I charcoal gas mask was employed by both the British and American. Activated charcoal was found to be one of the best adsorbents.

2. To create high vacuum in vessels, Tail and Dewar used activated charcoal.For dehydration and also purification of gases like CO_2, N_2, Cl_2, O_2 and He, alumina and silica are employed. In the blast furnace silica gel is also used for drying air.

3. One of the highly important use of adsorption is the softening of hardwater. Permutit is employed for this process which adsorbs Ca^{2+} and Mg^{2+} ions in its surface, there is an ion exchange as shown below it occurs on the surface.

$$Na_2Al_2Si_4O_{12} + CaCl_2 \rightarrow CaAl_2Si_4O_{12} + 2NaCl$$

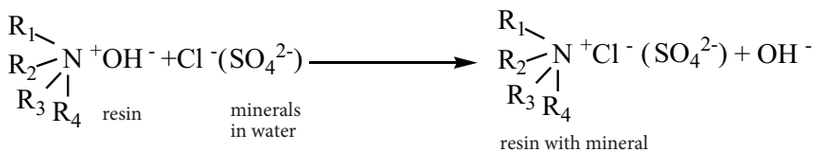
Exhausted permutit is regenerated by adding a solution of common salt.

$$CaAl_2Si_4O_{12} + 2NaCl \rightarrow Na_2Al_2Si_4O_{12} + CaCl_2$$

4. Ion exchange resins

Ion exchange resins are working only based on the process of adsorption. Ion exchange resins are used to demineralise water. This process is carried out by passing water through two columns of cation and anion exchange resins.

$$\underset{\text{resin}}{2\,RSO_3H} + \underset{\text{minerals in water}}{Ca^{2+}(Mg^{2+})} \rightarrow \underset{\text{resin with mineral}}{(RSO_3)_2Ca(Mg)} + 2H^+$$



5. Petroleum refining and refining of vegetable oil

Fuller's earth and silica gel are used for refining process.

6. Decolourisation of Sugar:

Sugar prepared from molasses is decolourised to remove coloured impurities by adding animal charcoal which acts as decolourising material.

7. Chromatography

The chromatographic technique is applied for separation of components in a mixture. It is mainly based on adsorption of components on the surface of adsorbents. This method is very effective and used for identification, detection and estimation of many substances even if they are contained in micro quantities.

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